AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 3 · STUDENT LESSON

Classification or Regression?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Define the target. Match the pairs. Keep the score's unit.

Distinguish category prediction from numerical estimation and explain why their checks can disagree.

Keep one habit: Target contract → duration/height contrasts → six-request clinic → guarded comparisons → independent print jobs.

MAIN LESSON · SECTION 1

What does the target value mean to the user?

> A number can encode a category or measure an amount; choose checks that match that meaning.

Your learning goals

I can identify a task from the meaning of its target.

I can distinguish category agreement from numerical error.

I can specify **output**s and checks that match a user need.

Recall prediction-time availability

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.
- After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.
- At dispatch, may the observed arrival duration be used as an **input** to predict that same trip duration? Explain.

Your end goal

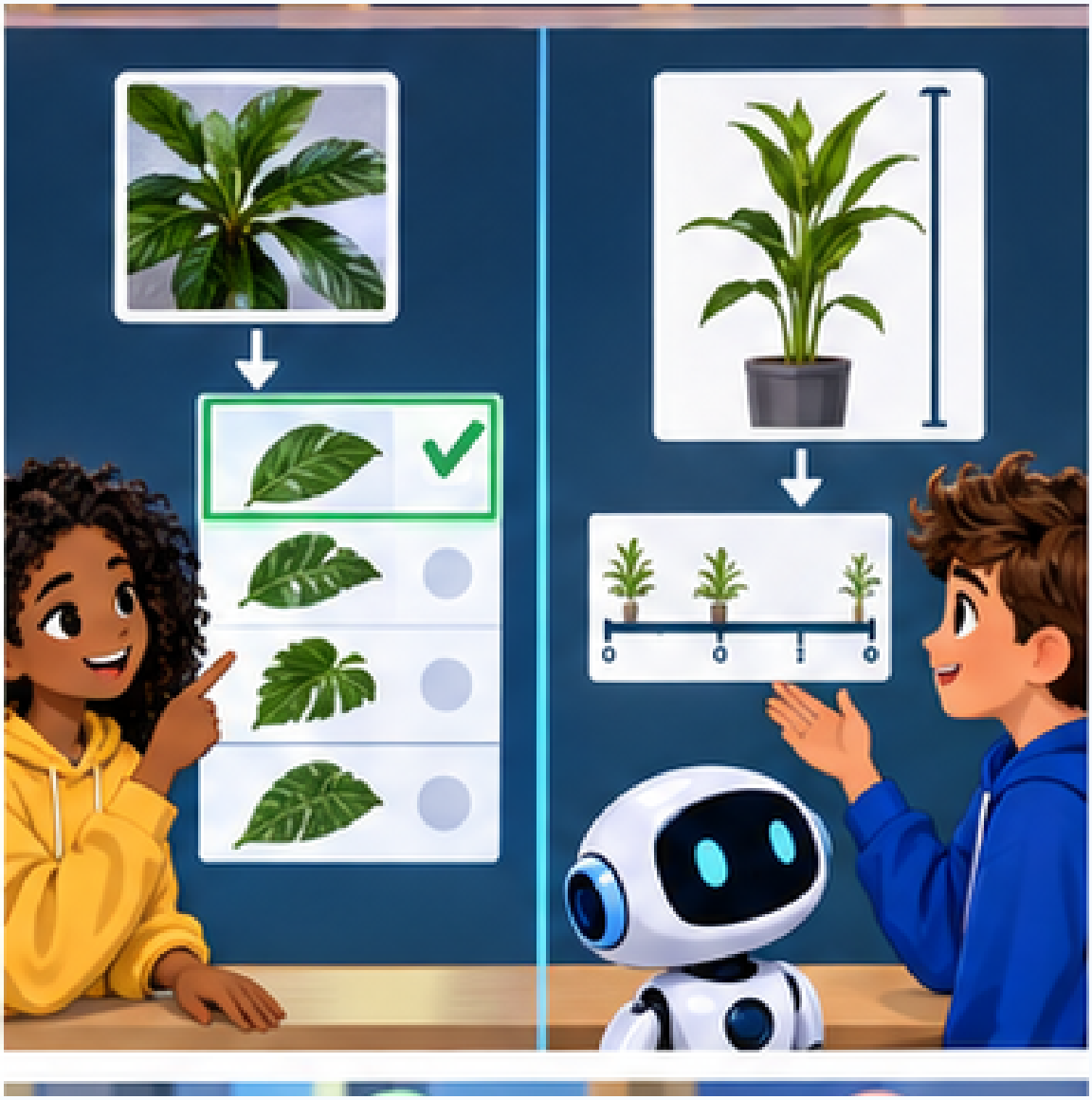
By the end, I can explain **classification** or **regression**? using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Classification predicts a category; **regression** predicts a numerical quantity. The target—not the **input** format—determines the task type.
</details>



LEARNING TARGET System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Ask for a category or ask for a quantity

> The same trip can support two different target questions.

Compare the dispatch-time questions

At robot dispatch, we can ask two questions about the same fictional trip. Will it take more than 10 minutes? Or how many minutes will it take? The first asks for a category. The second asks for an amount with units.

The target question determines the task, even if both models would use the same dispatch-time **input**s.

Question	Target meaning	Task
Will the trip exceed 10 minutes?	0 for no and 1 for yes	Classification
How long will the trip take?	Elapsed duration in minutes	Regression
Which route type will be used?	Indoor or outdoor	Classification

Use target meaning rather than decimal appearance

Classification predicts a category from a defined set. **Regression** estimates a numerical quantity for which differences have meaning.

A category may be coded with numbers: 0 and 1 here are names for no and yes, not travel times. Conversely, a duration recorded as a whole number of minutes can still be a **regression** target.

Decimal points do not decide the task.

Preserve the prediction-time boundary

Keep Week 2's **prediction** time: route length and the saved queue may exist at dispatch, while the observed duration is available later for **training** or evaluation.

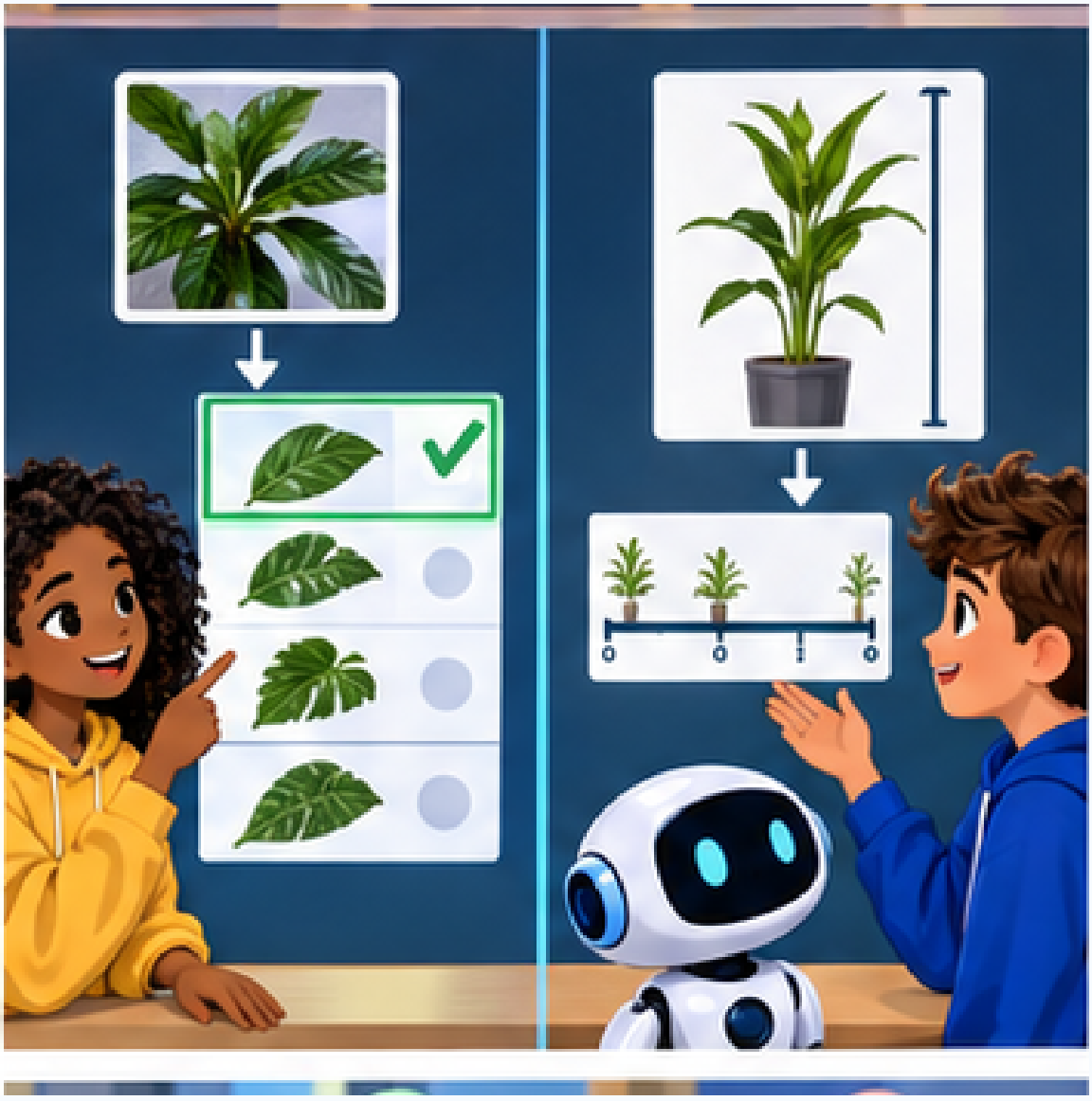
Do not feed that observed duration into an earlier dispatch **prediction**.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **classification** or **regression**? to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Compare category agreement and duration discrepancy separately

> The predictions are supplied examples; this lesson does not fit a model.

Read the target-and-check contract

Question	Target meaning	Output check
Will the trip exceed 10 minutes?	No/yes category; 0 and 1 are category codes	Compare category with the defined observed category
How many minutes will it take?	Numerical duration with meaningful differences	Absolute discrepancy in minutes; average error with case count
Can a duration estimate be thresholded?	Yes: this example uses estimate>10 to form a category	That does not turn a matching category into an exact duration estimate
Data pairing	Each actual/estimate pair must be the same trip	Equal lengths alone do not establish matching identities

The following **prediction**s are supplied for practice; no model is fitted here. Each estimate was written before revealing its matching observed duration.

We compare duration error with a category decision made from that estimate. Predict slow when the estimated duration is greater than 10 minutes; exactly 10 is not slow.

A **class**ifier could also predict categories directly, but this example uses a **threshold** on estimates.

Complete paired-duration comparison script

```
actual_minutes = [8, 10, 14]
estimated_minutes = [9, 11, 12]
correct = 0
errors = []
for actual, estimate in zip(actual_minutes, estimated_minutes):
    actual_slow = int(actual > 10)
    predicted_slow = int(estimate > 10)
    correct = correct + int(actual_slow == predicted_slow)
    errors.append(abs(estimate - actual))
print(correct, len(actual_minutes))
print(errors)
```

Trace errors, matching order and mean units

zip pairs the lists in order.

They must have equal lengths and refer to the same trips; otherwise this loop is not a valid evaluation. abs gives the distance from zero, so both a one-minute overestimate and a one-minute underestimate have **absolute error** 1.

The **output** is 2 3 followed by [1, 1, 2]. The mean **absolute error** is (1 + 1 + 2) / 3 = 4/3 minutes, about 1.33 minutes.

Read the three per-trip comparisons

Read both kinds of evidence

ACTUAL AND ESTIMATE	DURATION ERROR	CATEGORY MATCH
8 and 9 minutes	1 minute	No and no match
10 and 11 minutes	1 minute	No and yes differ
14 and 12 minutes	2 minutes	Yes and yes match

All three rows use the same greater-than-10 boundary. A small duration error can cross that boundary; a larger error can stay on the same side.

Explain the deciding check

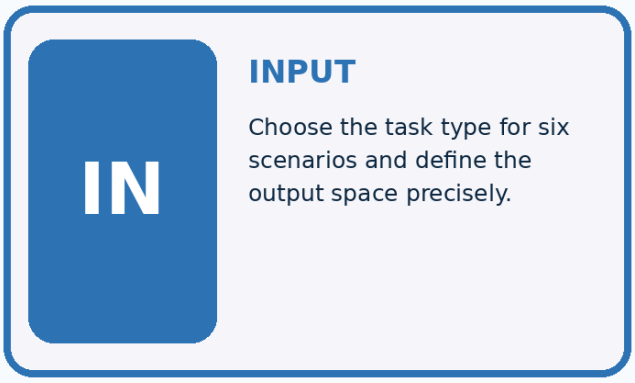
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

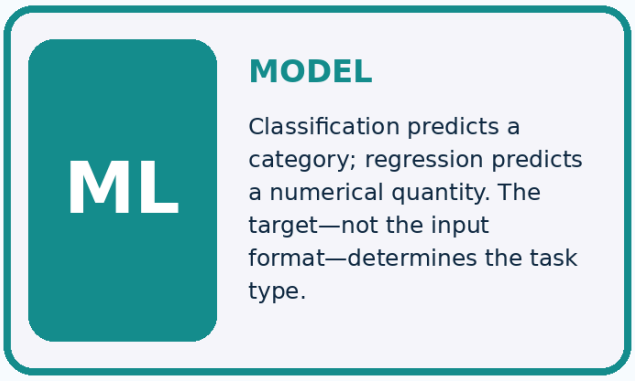
▼ **Earlier wording, alternatives and picture transcripts**

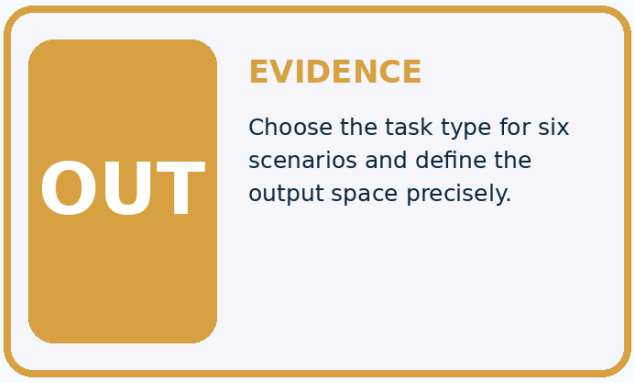
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Choose the task type for six scenarios and define the **output** space precisely.

MODEL IDEA **Classification** predicts a category; **regression** predicts a numerical quantity. The target—not the **input** format—determines the task type.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Predict plant species from a photo: target is a category, so **classification**. Predict plant height next week: target is centimeters, so **regression**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

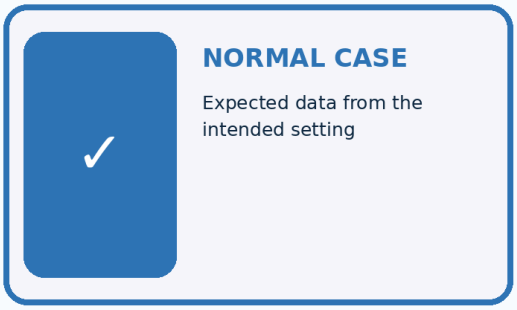
Connect the picture and the explanation

MAIN LESSON · SECTION 4

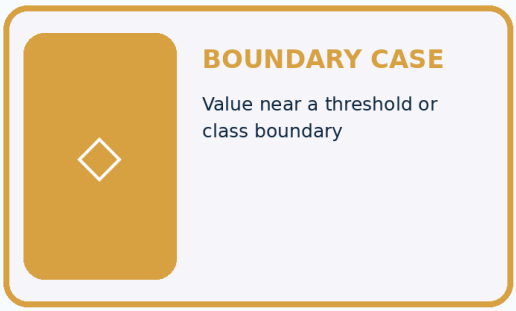
Compare a small quantity error with a threshold-category error

> A boundary can make two checks disagree about the same estimate.

Four evaluation conditions



NORMAL Expected data and expected behavior.

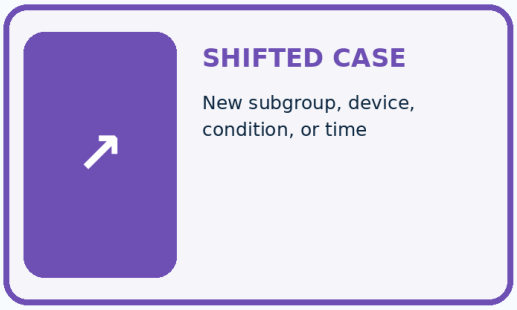


Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Change the category boundary to strictly greater than 7 cm.

▼ **Check after predicting**

Prediction: Actual and estimated categories both become short, short, tall: 3/3 correct.

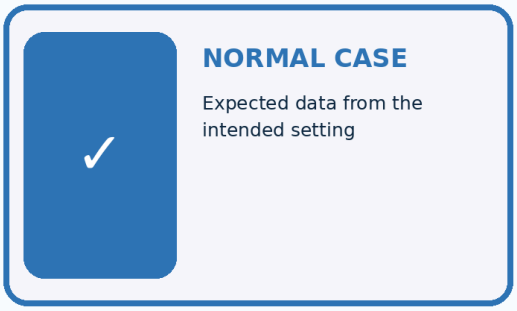
Reason: The numerical estimates did not improve; the decision definition changed.

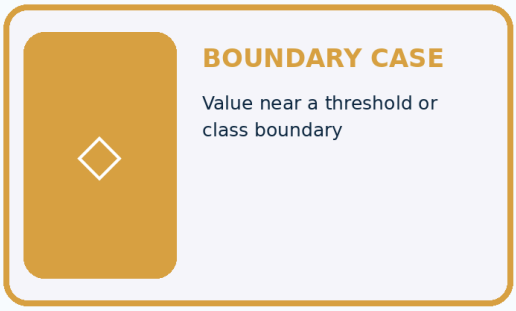
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

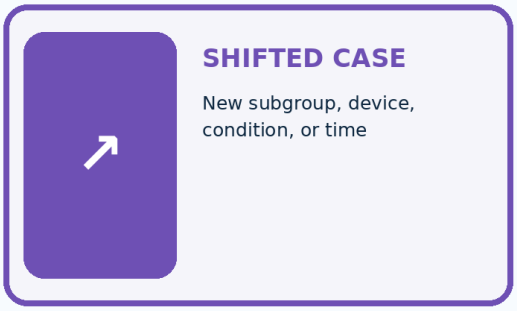
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the same height estimates under two category definitions

> The categorical score can change while numerical error stays fixed.

Read the complete height pairs and scope

Distinguish predicting growth height from predicting a growth category.

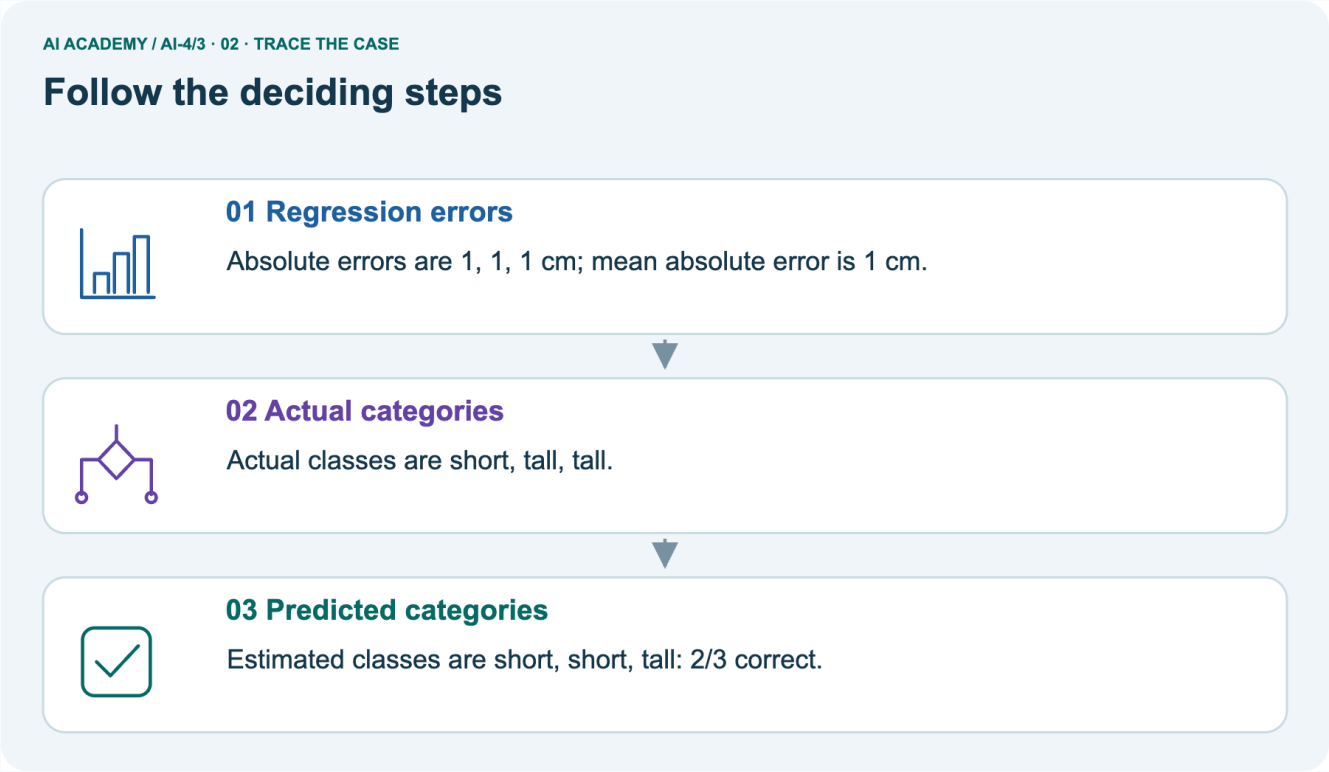
Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

Keep units and scenarios separate. Dispatch durations actual 8,10,14 and estimated 9,11,12 give errors 1,1,2 minutes and 2/3 categories correct at>10. Pictured heights actual 4,7,9 cm and estimates 5,6,10 cm give errors 1,1,1 cm. Strict >6 cm gives 2/3 matching categories; changing only to >7 cm gives 3/3. Height estimates and MAE 1 cm do not change.

These are measured-height targets, not growth increments; no initial-height differences are supplied. Changing the category definition is not evidence of a better numerical estimate.

Trace errors and categories



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Regression errors	Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.
2. Actual categories	Actual classes are short, tall, tall.
3. Predicted categories	Estimated classes are short, short, tall: 2/3 correct.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original paired-height graphic

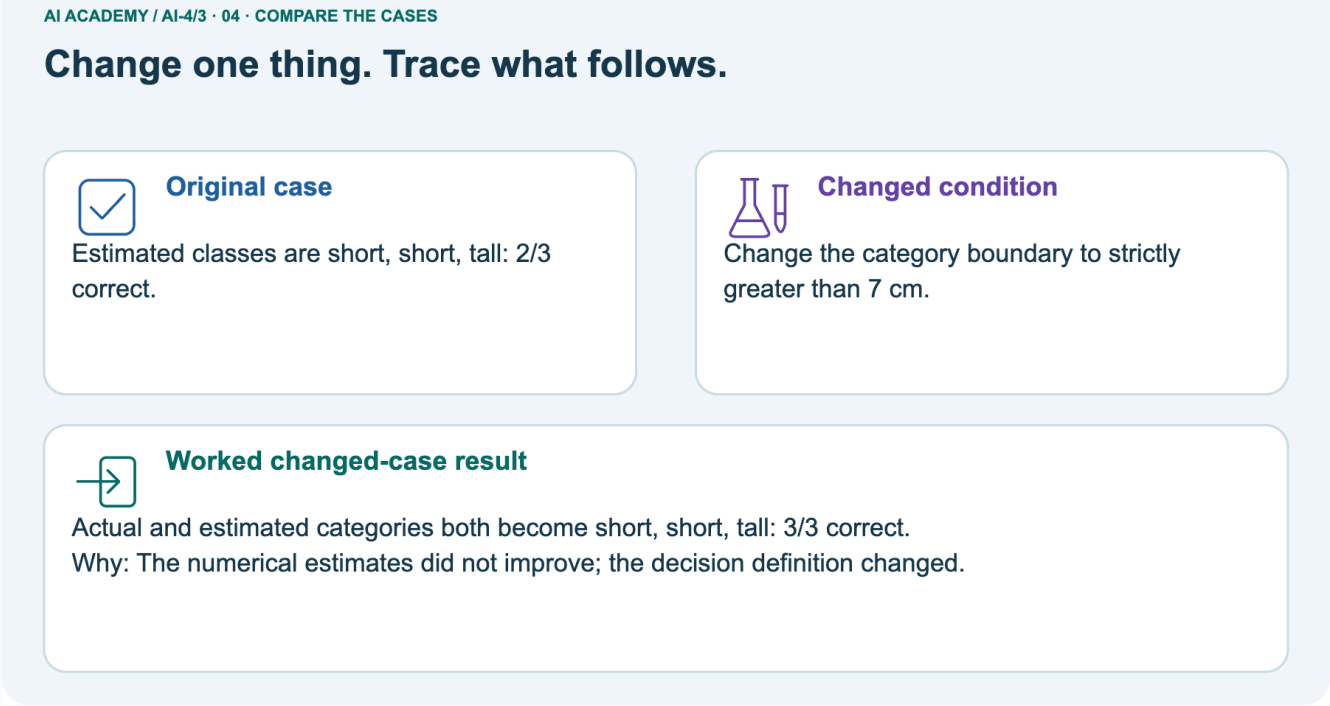


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict changing the strict boundary

- Change one thing: Change the category boundary to strictly greater than 7 cm. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State target and consequence limits

- **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

- Read the labelled picture: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Reference · Separate duration comparison

The 14-minute trip has the largest duration error in our example, yet its slow category is correct. The 10-minute trip has only a one-minute error but the category is wrong. These checks answer different questions.

Report 2 correct out of 3 category decisions separately from a mean **absolute error** of about 1.33 minutes. Do not compare a fraction directly with minutes.

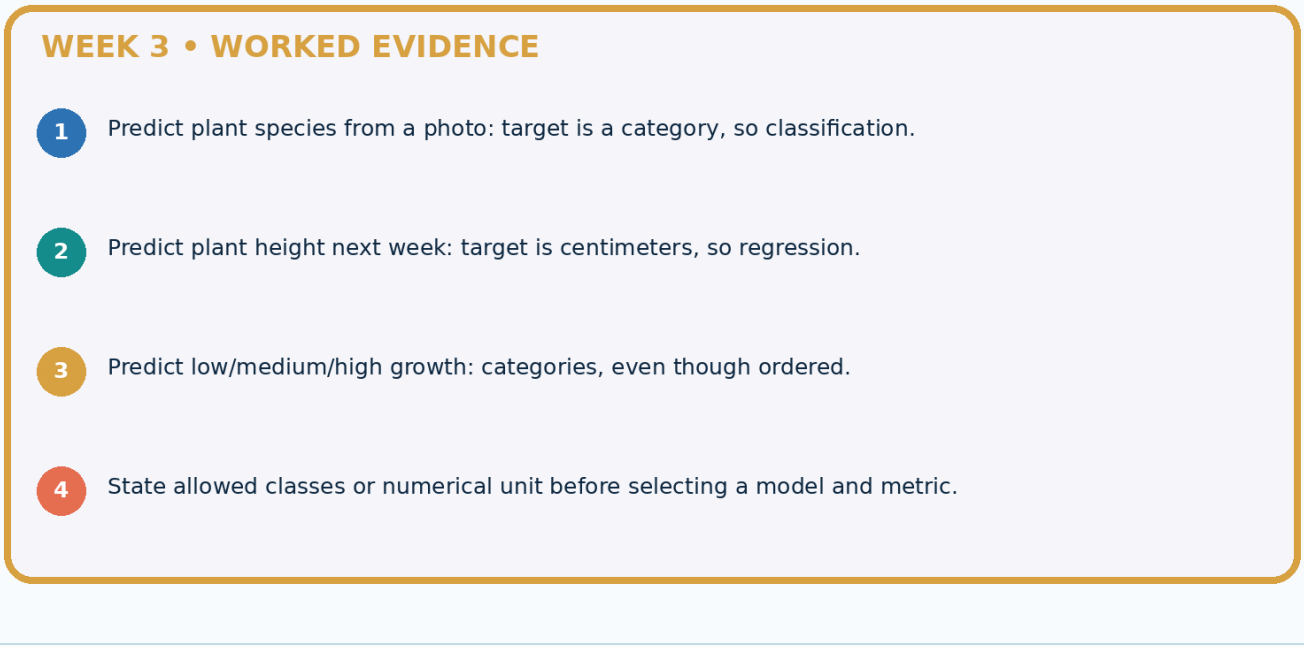
> **A trip actually takes 10 minutes and its estimate is 10.5 minutes. At the greater-than-10 boundary, give the actual category, predicted category and absolute duration error. Is the category decision correct?**

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Predict plant species from a photo: target is a category, so **classification**.

STEP 2 Predict plant height next week: target is centimeters, so **regression**.

STEP 3 Predict low/medium/high growth: categories, even though ordered.

STEP 4 State allowed **class**es or numerical unit before selecting a model and metric.

CHECK Verify the denominator, unit, split, **class** definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

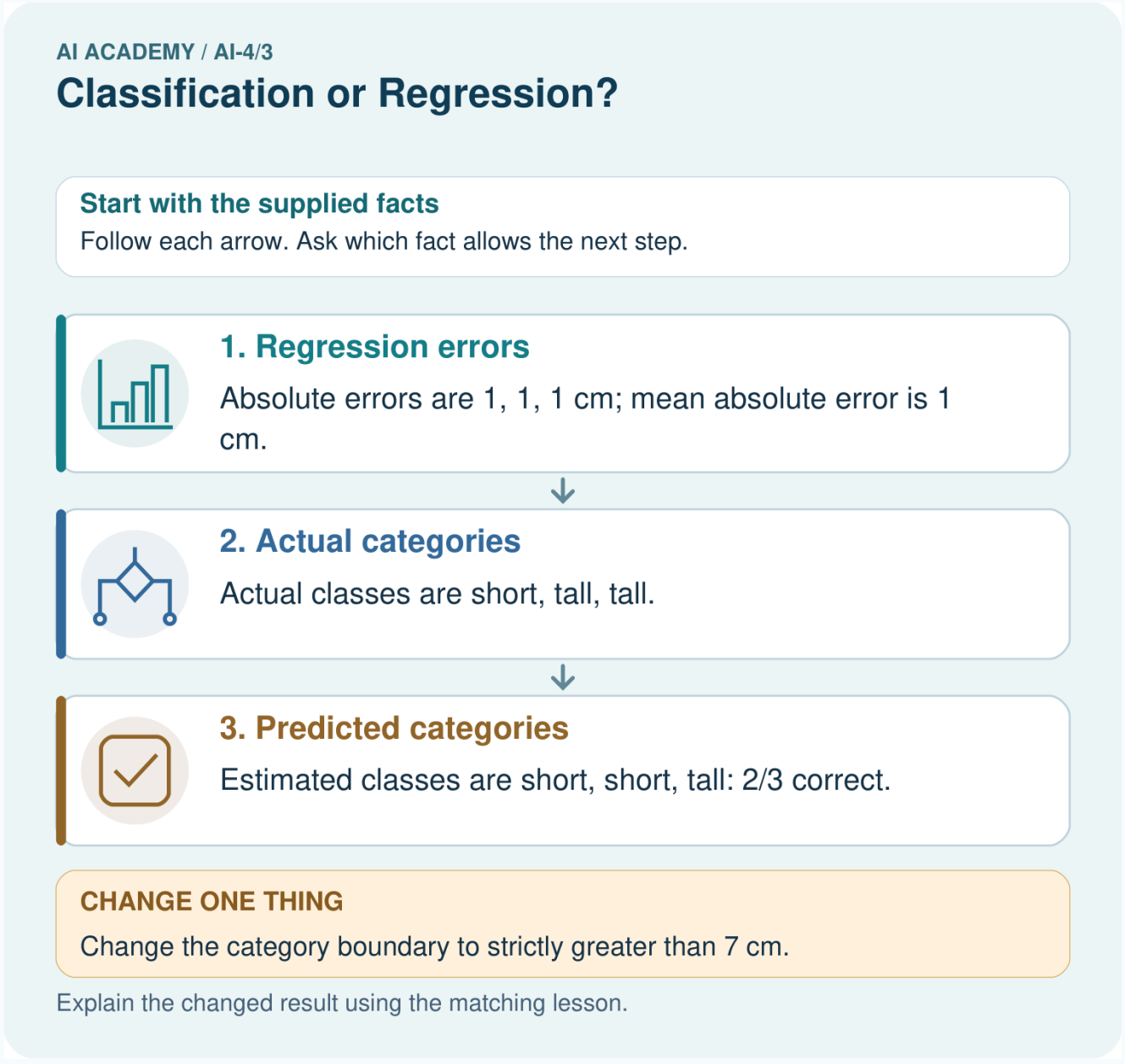
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Classification or Regression?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find category codes mistaken for measured quantities

> Ask what a difference between target numbers would mean.

Inspect the first-error table

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

State output meaning and limits

- A task specification should name the **prediction** moment, permitted **input**s, target meaning and how **prediction**s will be checked. **Classification** checks compare category decisions with observed categories.

  Regression checks measure numerical error with stated units. Later weeks examine additional metrics and why the consequences of different errors matter.

- A correct category does not imply an exact quantity, and low numerical error does not guarantee every **threshold** decision is right. Three supplied examples establish only these calculations.

  They do not establish a trained model's reliability or show that a robot should act without review.

Challenge metric mismatch

- Diagnose one cause and name the cleanest next experiment:

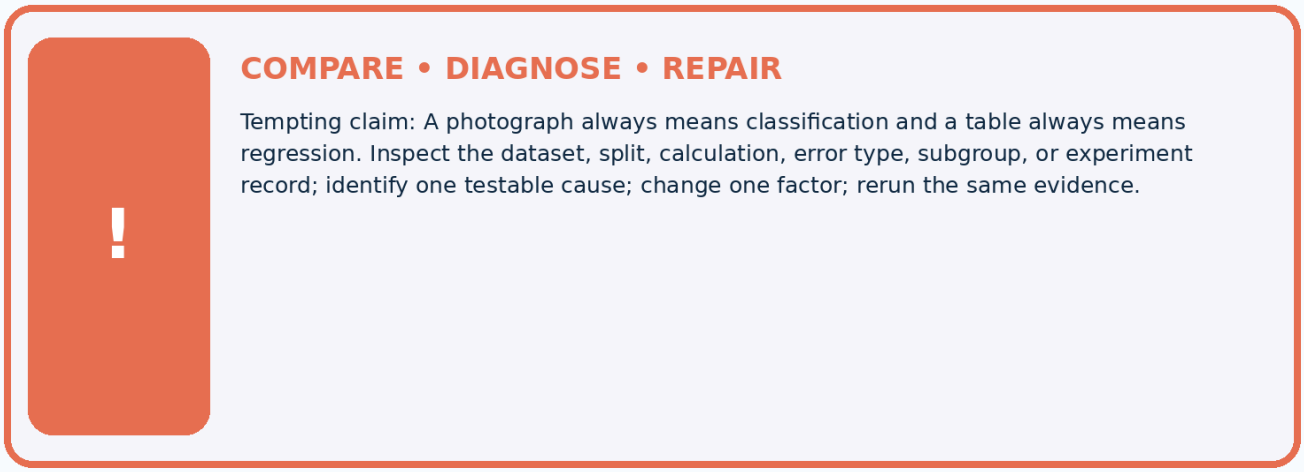
CLAIM TO CHALLENGE A photograph always means **classification** and a table always means **regression**.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



A closer explanation

> **A team needs an estimated arrival duration in minutes. State the task type and a suitable error check. Explain why reporting only the fraction of correct slow or not-slow decisions leaves its main question unanswered.**

MAIN LESSON · SECTION 8

Practise the 14-minute example with less help

> A matching slow label does not imply an exact duration.

Complete supplied W2

Use workbook W2 for the connected example "**Classification** and **regression**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit numbers-always-mean-regression reasoning

- Use the worked case for **Classification** and **regression**. Proposed explanation: "Numbers always mean **regression**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured unchanged MAE

- Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Keep independent print-job comparison separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Compare encoded categories and binned durations

A learner says that predicting route type is **regression** because indoor is encoded as 1 and outdoor as 2. The numbers are category codes: their difference does not measure travel time or another meaningful amount.

Treat this as **classification** and judge the predicted category. Changing the codes must not change the meaning of the task.

Another learner puts durations into short and long bins, then claims to predict exact minutes. Binning creates a **classification** target and loses within-bin detail.

If the user needs minutes, keep a quantity target and evaluate error in minutes. If the user needs a **threshold** decision, define the boundary and relevant consequences. Neither task is automatically better.

▼ **Reference · Experiment-record fields**

QUESTION + intended use + success **threshold**

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive **class** + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

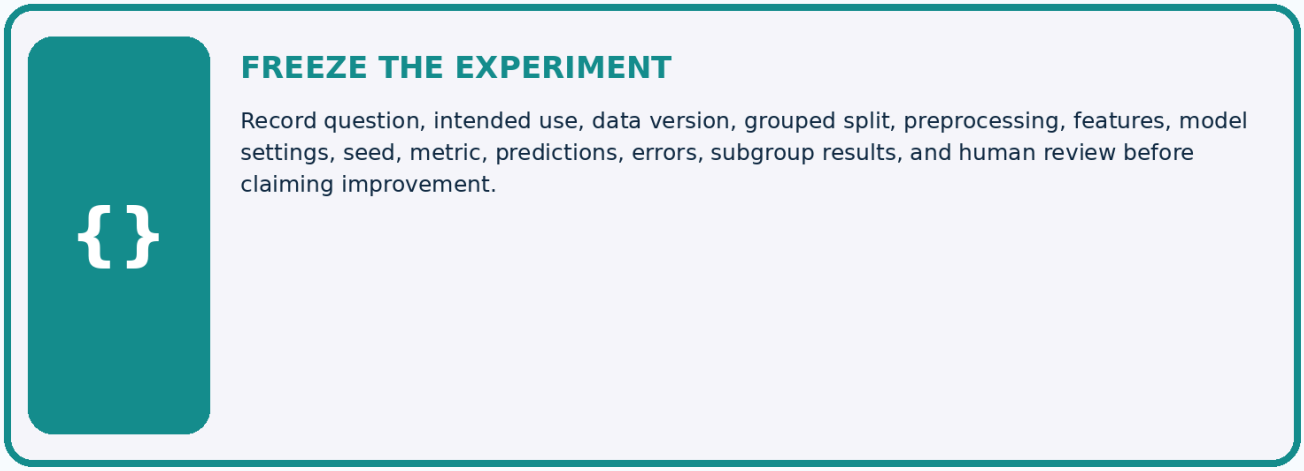
Write the minimum record needed to reproduce today's result:

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

A closer explanation

Practise with less help: Trace both checks

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a six-request clinic and four complete comparison probes

> Target definitions and measurement units decide which score is meaningful.

Six supplied problem statements

Complete fictional user request	Task/target and suitable check
1 · At capture, predict label healthy or spotted from a specified allowed set.	Classification; category agreement under a declared label rule. No real plant diagnosis is validated.
2 · Predict the plant's height at day 7 in cm, with numerical discrepancies meaningful.	Regression; absolute error/MAE in cm, with sample size and case identities.
3 · Predict indoor/outdoor route kind, encoded 1/2.	Classification; numbers are names for route categories, not travel duration.
4 · Estimate number of boxes delivered, judging how many boxes the estimate misses by.	Numerical quantity target; regression under this stated error meaning even if observations are whole counts.
5 · Predict whether the plant will grow well.	Underspecified: rewrite as height>6cm at day7(category), or height at day 7 in cm(quantity); state which user need matters.
6 · Predict small/medium/large class, encoded 0/1/2 under a supplied category rule.	Classification with ordered classes; code differences are not automatically cm differences.

Complete normal, boundary, missing and unit probes

Complete paired-data probe	Expected evidence
Normal heights: actual [4,7,9], estimated [5,6,10], strict >6 cm	MAE 1 cm, category 2/3
Boundary change: same height pairs, strict >7 cm	MAE still 1 cm, category 3/3; changed target definition
Missing pair: 3 actual heights but only 2 estimates	Reject incomplete pairing; do not silently truncate to report a three-case score
Unit mismatch: actual in cm, estimates in mm	Convert under an explicit unit contract before comparing; the numeric function cannot infer units

Specify, predict, compare and explain the user need

1. Task-Type Clinic: classify all six complete requests, rewrite the vague target and name the output meaning and check. For the height probes, predict category lists, count and numerical errors before the optional complete program. Its length/nonempty/numeric checks prevent some invalid calculations; they do not verify same-case identity, unit consistency or measurement accuracy.

   Original zip-based code assumes correctly paired equal-length lists; the new check rejects length mismatch rather than silently truncating. No model is trained here and estimates are supplied classroom values. Keep the selected print-job actual/estimate lists and strict 5-minute rule for the first independent attempt.

Complete optional guarded comparison program

```
import math

def compare_paired(actual, estimated, boundary):
    # Caller must supply matching cases in the same order and the same units.
    if len(actual) != len(estimated) or not actual:
        raise ValueError('nonempty equal-length paired values required')
```

```
    for value in list(actual) + list(estimated) + [boundary]:
        if type(value) not in (int, float) or not math.isfinite(value) or value < 0:
            raise ValueError('finite nonnegative numeric values required')
    actual_classes = [int(value > boundary) for value in actual]
    predicted_classes = [int(value > boundary) for value in estimated]
    errors = [abs(pred - ref) for ref, pred in zip(actual, estimated)]
    correct = sum(ref == pred for ref, pred in zip(actual_classes, predicted_classes))
    return {'actual_classes': actual_classes,
            'predicted_classes': predicted_classes,
            'correct': correct, 'cases': len(actual),
            'absolute_errors': errors, 'mae': sum(errors) / len(errors)}

actual = [4, 7, 9]
estimated = [5, 6, 10]
for boundary in [6, 7]:
    result = compare_paired(actual, estimated, boundary)
    print('boundary', boundary, 'cm')
    print('classes', result['actual_classes'], result['predicted_classes'])
    print('correct', result['correct'], 'of', result['cases'])
    print('MAE', round(result['mae'], 2), 'cm')
```

Keep the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Task-Type Clinic: classify six problem statements, rewrite vague targets, and match each to a useful metric.

1. Write the hypothesis and expected **evidence** before running.

2. Task-Type Clinic: **class**ify six problem statements, rewrite vague targets, and match each to a useful metric.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the fresh print-job quantities and categories

> Explain what each numerical result measures rather than choosing one headline.

Independent paired records and scope

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

Pair the selected actual and estimated completion times in the stated order, using minutes throughout. Apply this task's own strict greater-than boundary, keep the per-case absolute errors and report the count before any fraction or mean. No model-fitting or broad performance evidence is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Compare new evidence independently

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is **classification** and which is **regression**?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's target and metric connected

> A feature table alone does not specify what a model should produce.

This week's contribution

For the Responsible Plant Classifier, define the target as a category set or a numerical quantity with units and observation time. Preserve pairing IDs, output interpretation and the metric denominator/unit. If numerical estimates are converted to categories, record that threshold as a separate decision definition. Do not call a changed threshold score an improvement in unchanged numerical predictions.

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain checkpoint, limits and human decision

THIS WEEK'S CHECKPOINT Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this **class**room prototype for medical, pesticide, agricultural, or other consequential decisions.

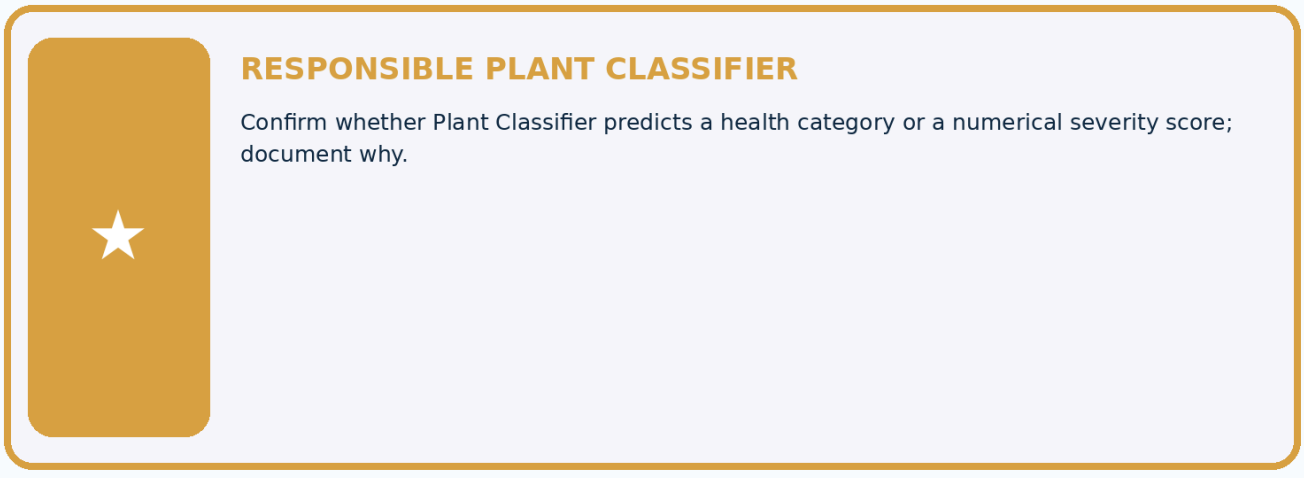
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish: state target meaning, score unit and comparison condition

> A category code and a measured amount answer different questions.

Explain without notes

- Close your notes. Explain this goal in your own words: Identify a task from the meaning of its target

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
classification	predict a category	
regression	predict a number	
class	one allowed category	
continuous target	number that can vary across a range	

Closer-explanation terms

Word	Meaning
Classification	Predicting a category from a defined set
Regression	Estimating a numerical quantity with meaningful differences
Threshold	A boundary used to turn a value into a decision

Word	Meaning
Absolute error	The nonnegative distance between an estimate and an observation

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

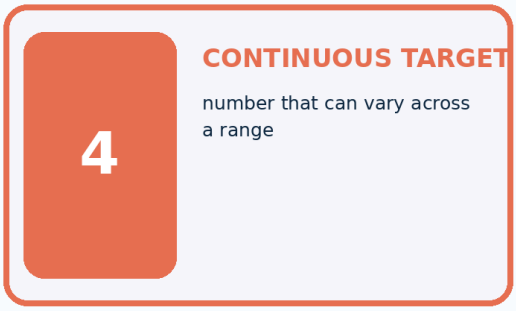
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A photograph always means **classification** and a table always means **regression**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **classification**, **regression**, **class**, **continuous target**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A photograph always means **classification** and a table always means **regression**.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

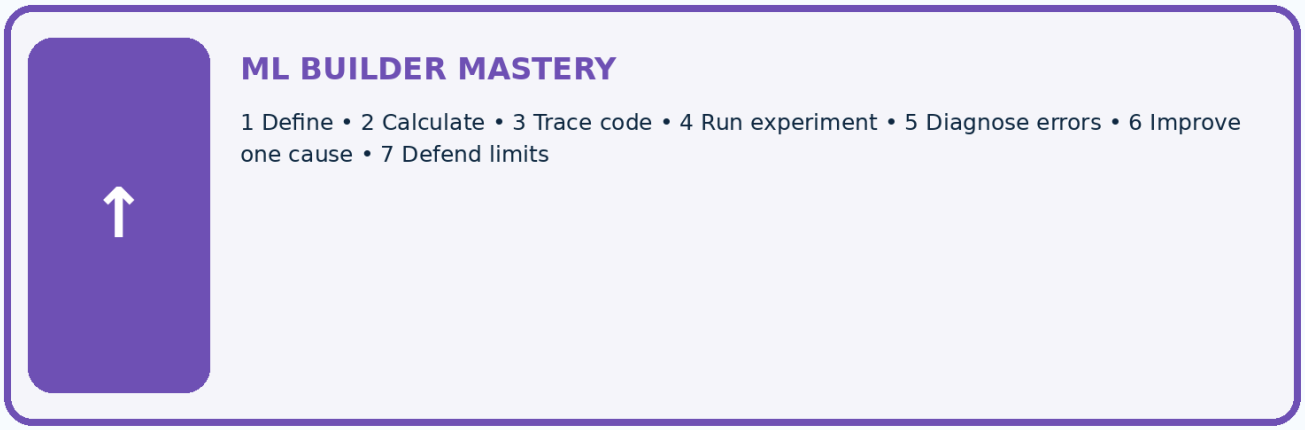
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep Compare the new print-job evidence independently for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

- Independent **evidence** and defense:

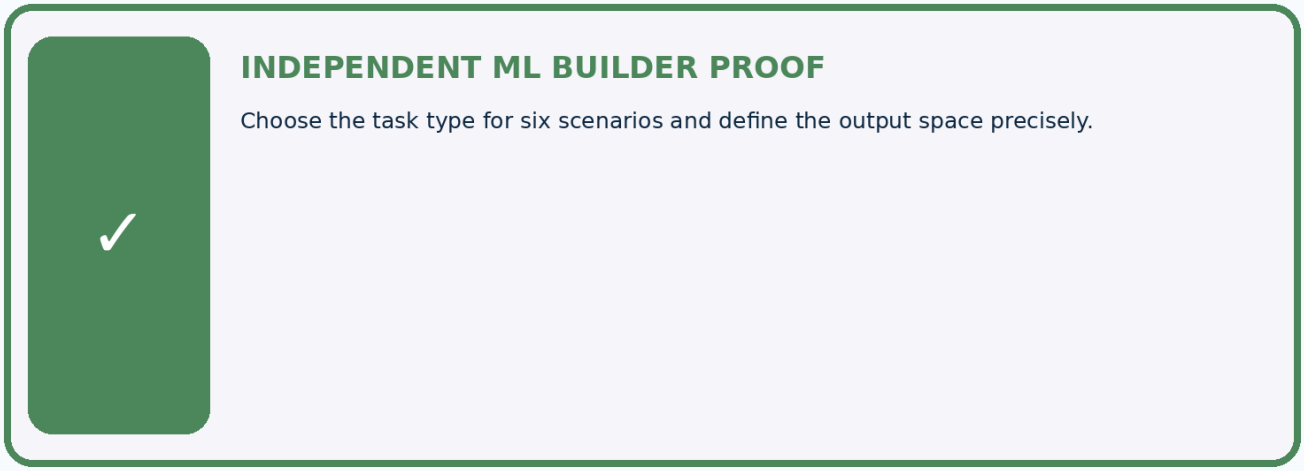
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Choose the task type for six scenarios and define the **output** space precisely.

NEXT BRIDGE Carry the experiment **evidence** into: Python and Table Readiness.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and **output** types.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

A is **classification** with categorical **output**; B is **regression** with numeric **output** and units.
Numeric **class** codes do not make **classification regression**.

Evidence or stage	Value or result
Healthy or spotted	Classification, category
Height next week	Regression, numeric cm
Class code 0 or 1	Still a category

▼ Earlier wording, alternatives and picture transcripts

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Choose the task by the meaning of the output

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and **output** types.

A is **classification** with categorical **output**; B is **regression** with numeric **output** and units. Numeric **class** codes do not make **classification regression**.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Identify a task from the meaning of its target	Explain the deciding step, show a trace or test, and state one limit.
Distinguish category agreement from numerical error	Explain the deciding step, show a trace or test, and state one limit.
Specify outputs and checks that match a user need	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several **class**es.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Distinguish predicting growth height from predicting a growth category.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Distinguish predicting growth height from predicting a growth category.

- Which labelled step matches this detail? Estimated **class**es are short, short, tall: 2/3 correct.
- 1. Actual categories
- 2. **Regression** errors
- 3. Predicted categories
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Predicted categories Estimated **class**es are short, short, tall: 2/3 correct. Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Change the category boundary to strictly greater than 7 cm. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Actual and estimated categories both become short, short, tall: 3/3 correct.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Repair and keep the record

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

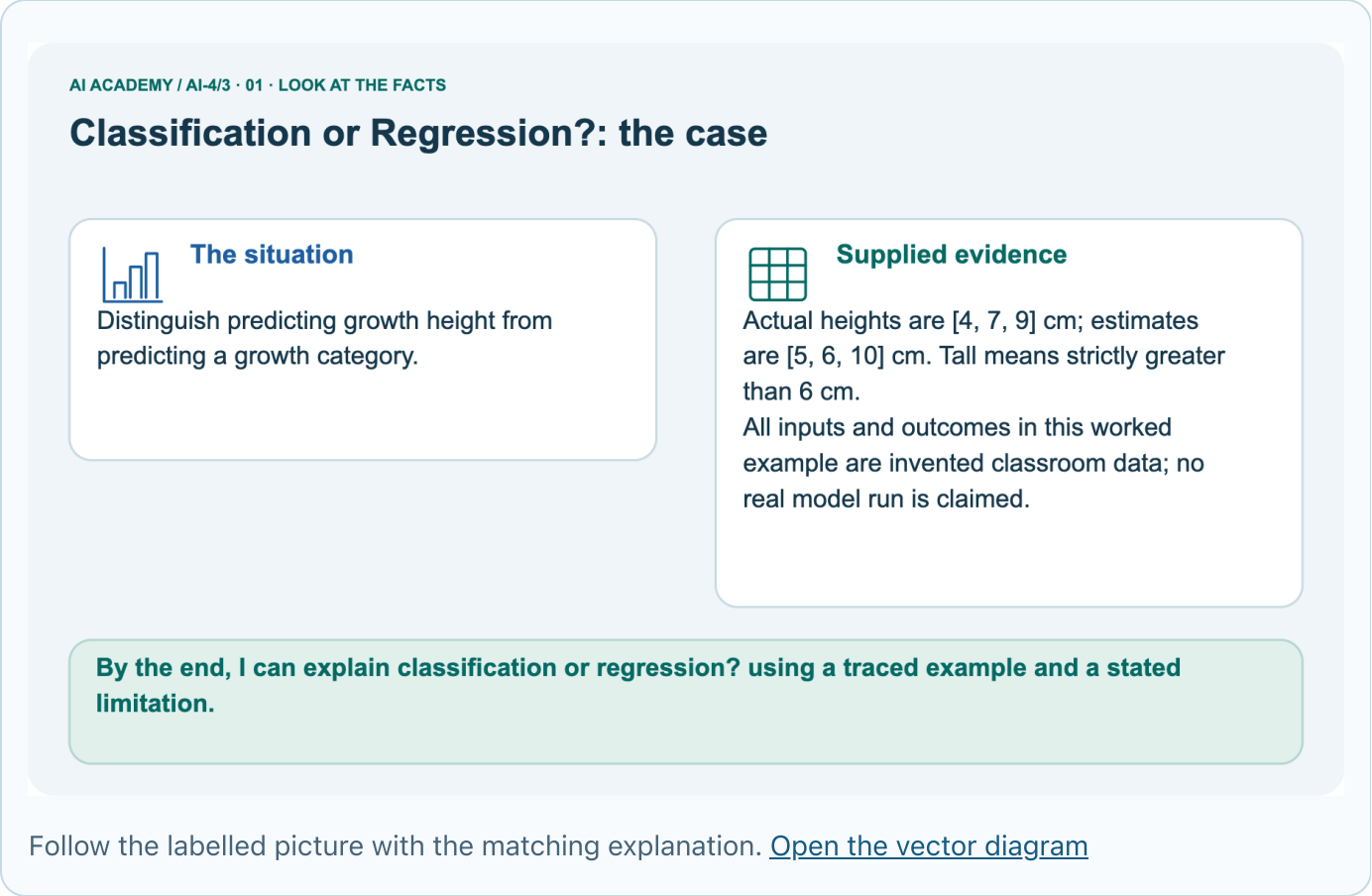
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Classification or Regression?: the case



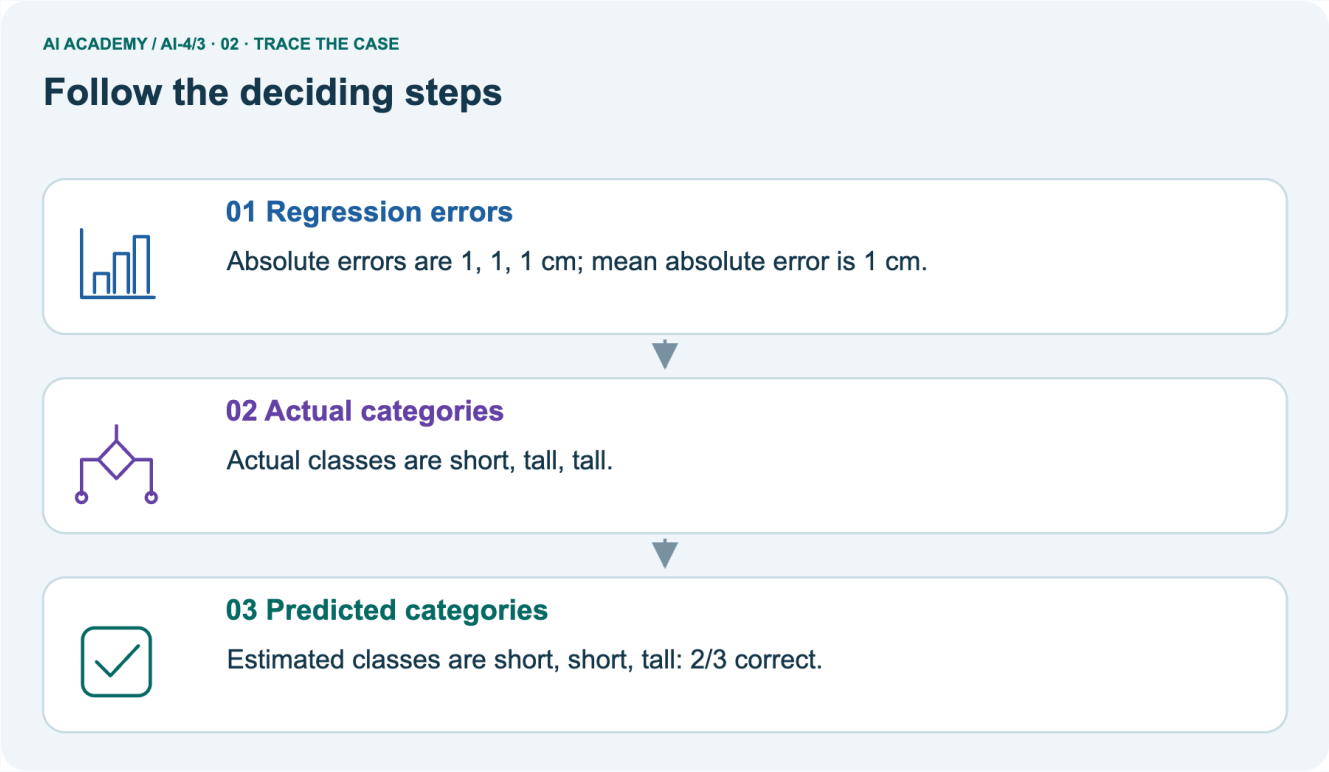
Follow the labelled picture with the matching explanation. Open the vector diagram

Distinguish predicting growth height from predicting a growth category. Supplied facts: Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regression errors: **Absolute error**s are 1, 1, 1 cm; mean **absolute error** is 1 cm. → Actual categories: Actual **class**es are short, tall, tall. → Predicted categories: Estimated **class**es are short, short, tall: 2/3 correct.

Point to each numbered stage and explain which recorded **input** it uses.

03. Actual and estimated heights (paired order)

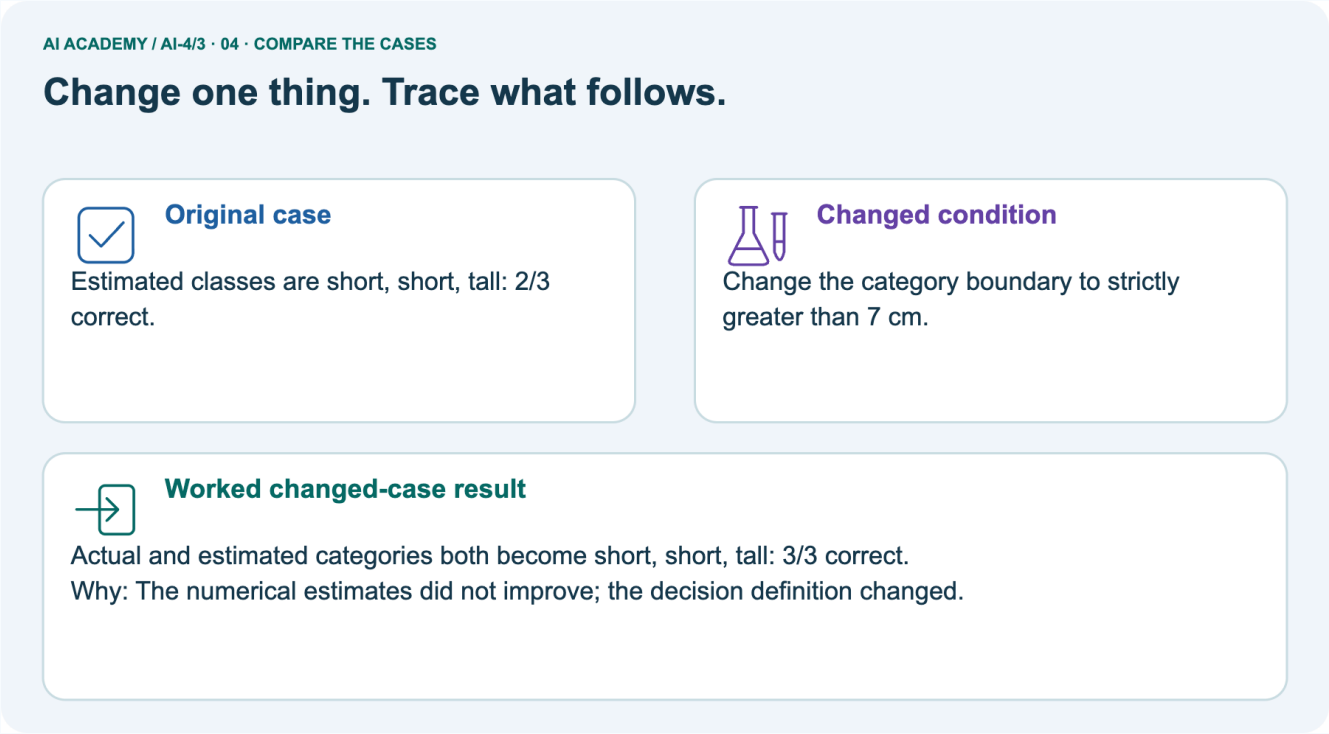


Follow the labelled picture with the matching explanation. Open the vector diagram

Actual and estimated heights (paired order). Units: cm. actual1: 4. estimate1: 5. actual2: 7. estimate2: 6. actual3: 9. estimate3: 10.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



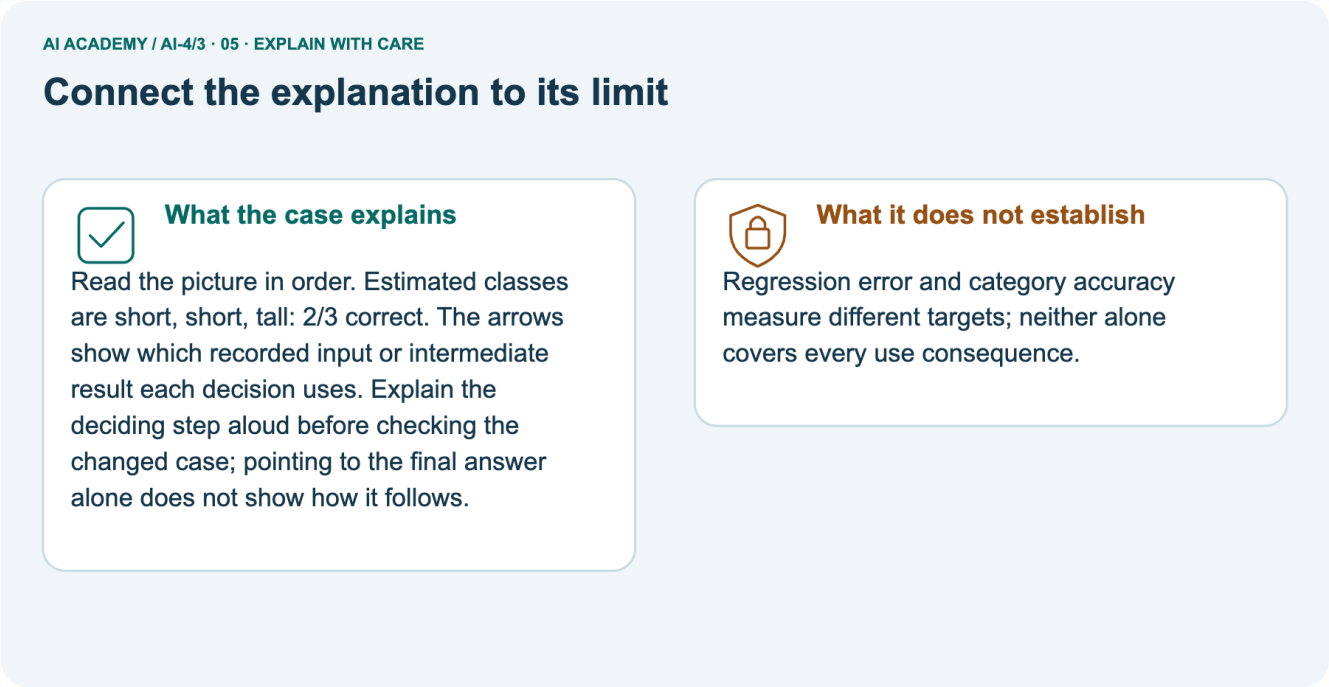
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Estimated **class**es are short, short, tall: 2/3 correct. Changed condition: Change the category boundary to strictly greater than 7 cm.

Expected result: Actual and estimated categories both become short, short, tall: 3/3 correct. Reason: The numerical estimates did not improve; the decision definition changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

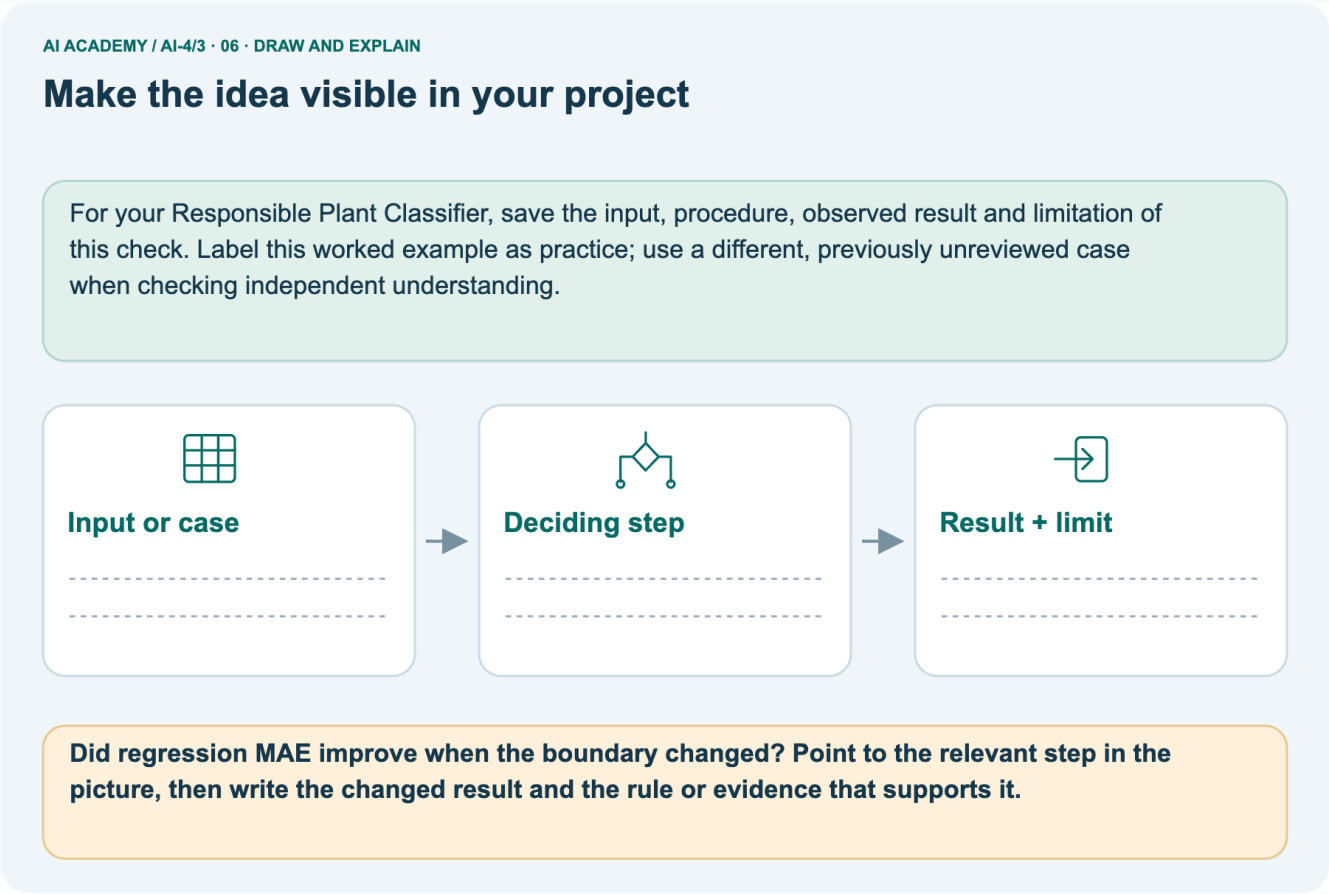
Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Classification or Regression?: the case

02. Follow the deciding steps

03. Actual and estimated heights (paired order)

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.